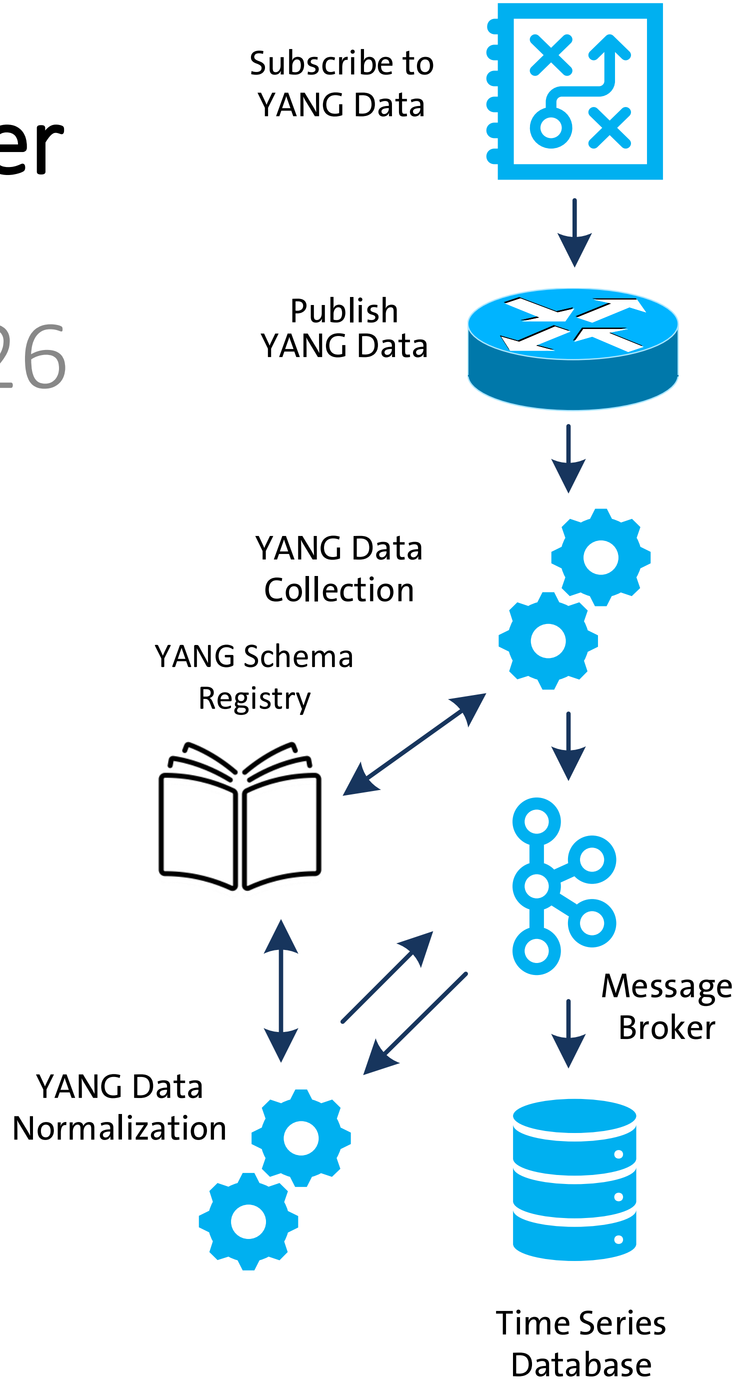
Validate YANG-Push to Message Broker End-To-End Data Processing Chain

IETF 126 Hackathon, July 18/19th 2026



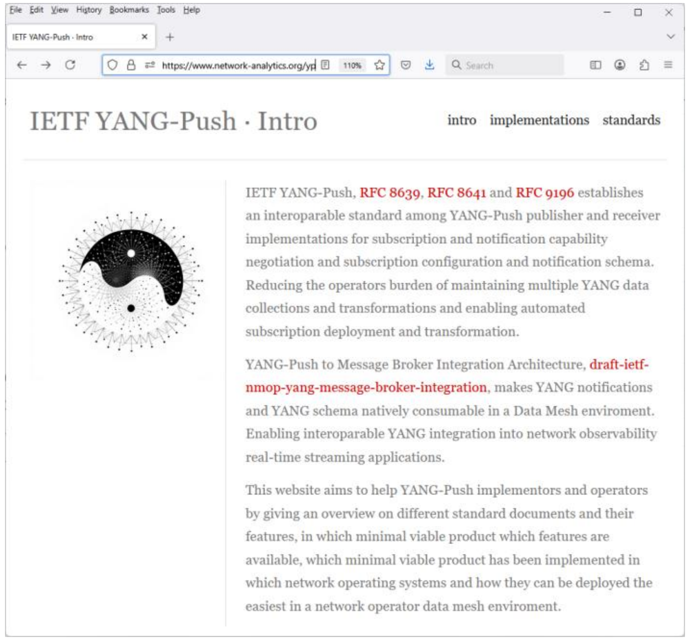
Hackathon Plan, Software and Website

Test Plan

Validate and verify

- 5 YANG-Push Publishers
- 2 YANG-Push Receivers
- 2 YANG-Push Network Telemetry Message
- 1 YANG Message Broker Producer and Schema Registry
- 3 YANG Message Broker Consumers

implementation in YANG data integration automation. Subscribe to YANG data on YANG-Publisher, obtain and register all YANG modules necessary to build YANG schema tree, register YANG schemas to Schema Registry and verify YANG notifications against scheme trees and produce and consume from Message Broker.



Software

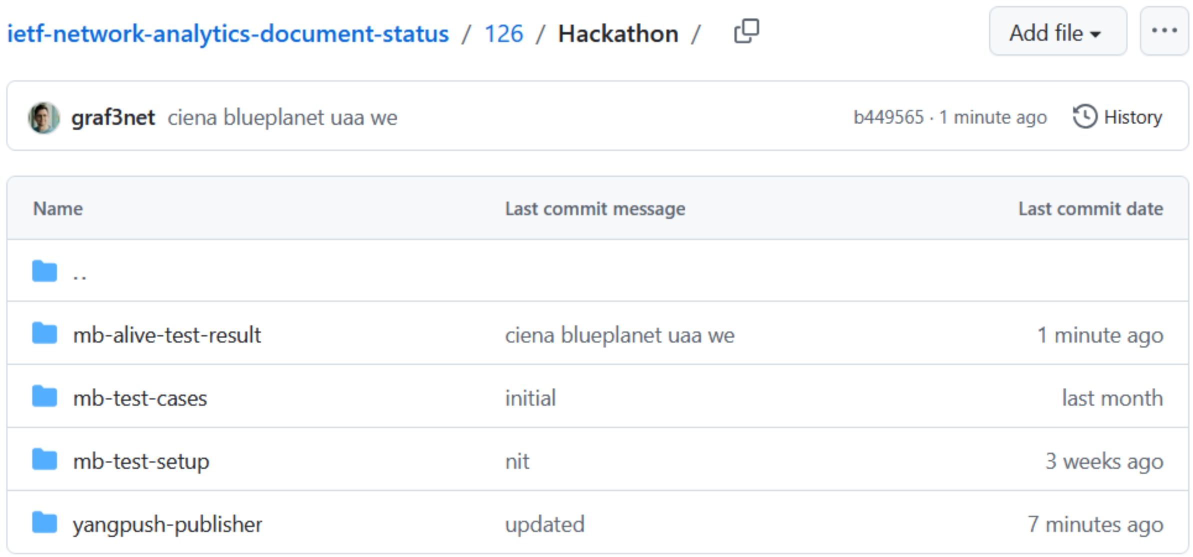
https://www.network-analytics.org/yp/how-to-deploy.html

- YANG-Push Publisher - Cisco IOS XR, 6WIND VSR, Huawei NE (Router),MA (OLT), Arrcus Arcos and Ductus DMAP
- YANG-Push Receiver – NetGauze™ and Pmacct
- YANG Message Broker Producer – NetGauze™
- YANG Message Broker Consumer – NetGauze™, Ciena Blue Planet UAA Workflow Engine, Cisco Crosswork Ingestion Service
- YANG Schema Registry – Apache Kafka YANG Plugin
- Libyang - https://github.com/CESNET/libyang/releases/tag/v5.4.9
- Yangkit - https://github.com/yang-central/yangkit/
- Wireshark - udp-notif dissector

Hackathon – Repositories

The results of the end-to-end tests

- Netconf RPCs and YANG-Push JSON and CBOR encoded notifications with wire captures from receiver
- YANG-Push Receiver cache content generation with YANG library context, YANG modules, metadata and notification schema validation logs
- YANG Message Broker Producer serialization logs
- YANG Schema Registry schema examples
- YANG Message Broker Consumer, serialization logs, YANG library context, YANG modules and message schema validation



https://github.com/network-analytics/ietf-network-analytics-document-status/tree/main/126/Hackathon/

An Architecture for YANG-Push to Message Broker Integration

draft-ietf-nmop-yang-message-broker-integration
draft-ietf-nmop-message-broker-telemetry-message
draft-ietf-nmop-yang-message-broker-message-key
draft-netana-nmop-message-broker-bmp-telemetry-message

- Subscription to YANG Notifications
  RFC 8639
- Subscription to YANG Notifications for Datastore Updates
  RFC 8641
- UDP-based Transport for Configured Subscriptions
  draft-ietf-netconf-udp-notif
- Subscription to Distributed Notifications
  draft-ietf-netconf-distributed-notif
- Extensible YANG Model for YANG-Push Notifications
  draft-ietf-netconf-notif-envelope
- Support of Versioning in YANG Notifications Subscription
  draft-ietf-netconf-yang-notifications-versioning
- YANG Modules Describing Capabilities for Systems and Datastore Update Notifications
  RFC 9196
- YANG Notification Transport Capabilities
  draft-ietf-netconf-yp-transport-capabilities
- YANG Library
  RFC 8525
- Augmented-by Addition into the IETF-YANG-Library
  draft-ietf-netconf-yang-library-augmentation
- Validating anydata in YANG Library context
  draft-netana-nmop-yang-anydata-validation
- Encoding of Data Modeled with YANG in the CBOR
  RFC 9254

```
   +------------------------------------------------------------+
   |                       Data Storage                         |
   +------------------------------------------------------------+
                                  ^
                                  | (12) Ingest Data
                                  | According to Schema
   +------------------------------------------------------------+
   |                       Data Consumer                        |
   +------------------------------------------------------------+
         |  ^                                   ^
(10) Get |  |                                   | (9) Validate
  Schema |  |                                   | Serialized Message
         |  |                                   | Schema on Consumer
         |  |                              +--------------------+
         |  |                              |      Message       |
         |  |                              |      Broker        |
         |  |                              +--------------------+
         |  |                                   ^
         |  |                                   | (8) Serialize
         |  | (11) Issue                        | YANG-Push Message
         |  |      Schema                       | annotated Schema
         v  |                  (6) Post         | ID on Producer
   +--------------------+          Schema  +--------------------+
   |        YANG        | <--------------- |  Data Collection   |
   |  Schema Registry   | ---------------> | YANG-Push Receiver |
   +--------------------+  (7) Issue       +--------------------+
         |                     Schema ID        |  ^ (3) Receive
         |                                      |  | YANG-Push
         v                         (4) Discover |  | Subscription
   +--------------------+                Schema |  | Start Message
   |       Stream       |          Dependencies |  |   ^
   |       Catalog      |            and Obtain |  |   | (5) Publish
   +--------------------+               Schemas |  |   | YANG-Push
                                                |  |   | Message
                                                |  |   | with
                          (1) Discover          |  |   | Subscriber
                              Notification      v  |   | ID
   +--------------------+     Capabilities +--------------------+
   |  Manage YANG-Push  | ---------------> |    Network Node    |
   |    Subscription    | ---------------> | YANG-Push Publisher|
   +--------------------+ (2) Subscribe    +--------------------+
```

YANG-Push Publisher - Subscription

- Netconf RPC calls to create, delete and obtain YANG-Push subscriptions and receiver configurations.

Name	Last commit message	Last commit date
..		
edit-config-yp-subscription-receiver-add.xml	testcases updated	last month
edit-config-yp-subscription-receiver-delete.xml	testcases updated	last month
get-yp-subscription.xml	testcases updated	last month
readme.md	testcases updated	last month

readme.md

Netconf Interactive Login

```
netconf-console2 --host=10.190.64.79 --port=830 --db=running -i --user=daisy1 --privKeyFile=/etc/daisy/net
```

Netconf Get YANG-Push subscriptions

```
netconf-console2 --host=10.190.64.79 --port=830 --db=running --user=daisy1 --privKeyFile=/etc/daisy/netgau
```

Netconf Create YANG-Push subscriptions with receiver

```
netconf-console2 --host=10.190.64.79 --port=830 --db=running --user=daisy1 --privKeyFile=/etc/daisy/netgau
```

Netconf Delete YANG-Push subscriptions with receiver

```
netconf-console2 --host=10.190.64.79 --port=830 --db=running --user=daisy1 --privKeyFile=/etc/daisy/netgau
```

YANG-Push Publisher - Notifications

- Subscription state change notifications as JSON and PCAP.
- Push update and change-update notifications as JSON and PCAP.

Name	Last commit message	Last commit date
..		
1-push-update.json	testcases updated	last month
1-subscription-started.json	testcases updated	last month
1-yp-subscripton.pcap	testcases updated	last month
2-push-update.json	testcases updated	last month
2-subscription-started.json	testcases updated	last month
2-yp-subscripton.pcap	testcases updated	last month
27-push-update.json	testcases updated	last month
27-subscription-started.json	testcases updated	last month
27-yp-subscripton.pcap	testcases updated	last month
readme.md	testcases updated	last month

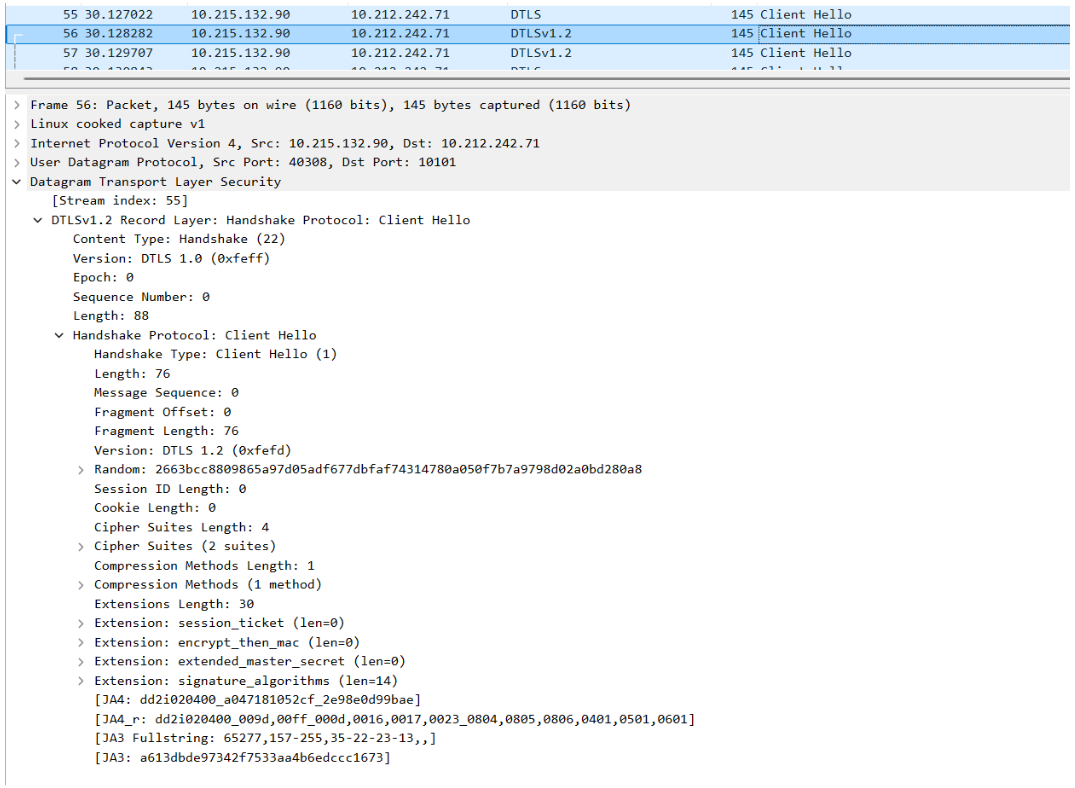
readme.md

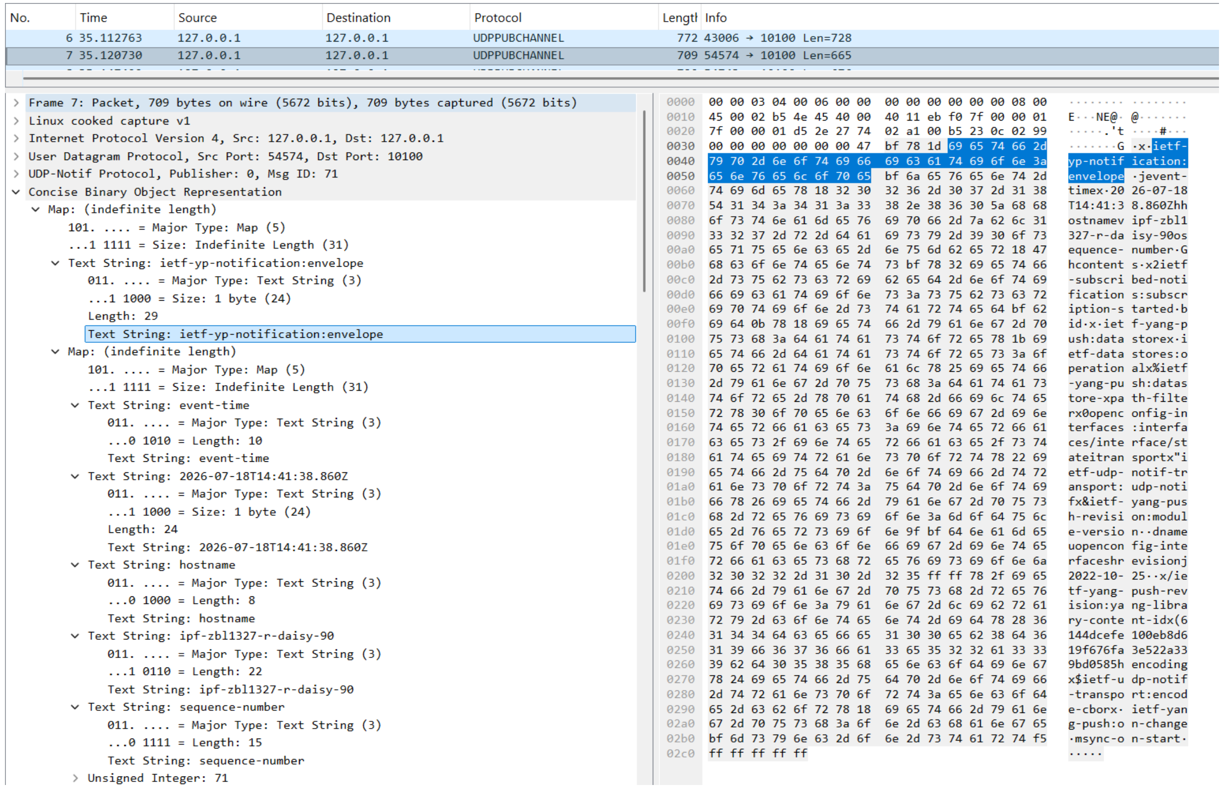
YANG-Push Notifications

subscription-started, push-update and push-change-update notifications in raw and json sorted by subscription id. pcap requires https://github.com/network-analytics/udp-notif-dissector Wireshark dissector.

YANG-Push UDP-Notif – DTLS and CBOR

- With Cisco IOS XR, 6Wind VSR and Arrcus Arcos we have now three implementations support [CBOR Name Identifiers]
- With Cisco IOS XR we have the first implementation support [DTLS 1.2 in UDP-Notif].





YANG-Push Capabilities & YANG Library - RPC

- Netconf XML encoded reply for RFC 9196 YANG-Push capabilities.
- Netconf XML encoded reply for RFC 8525 with draft-ietf-netconf-yang-library-augmentedby.

graf3net update bf0be62 · 16 hours ago History

Name	Last commit message	Last commit date
..		
get-systems-capa.xml	update	16 hours ago
get-yang-library.xml	update	16 hours ago

YANG-Push

IETF 126 Publisher Implementation Status

	6WIND VSR	Huawei NE	Huawei MA	Cisco IOS XR	Arrcus Arcos	Open-Source	Ductus DMAP
RFC 8639 YANG-Push Subscription	✓	✓	✓	✓	✓		✓
RFC 8641 YANG-Push Notification	✓	✓	✓	✓	✓		✓
draft-ietf-netconf-udp-notif	✓	✓	✓	✓	✓	✓	
draft-ietf-netconf-udp-notif with DTLS				✓			
draft-ietf-netconf-distributed-notif	✓	✓	✓		na		
draft-ietf-netconf-notif-envelope	✓	✓	✓	✓	✓		P
draft-ietf-netconf-yang-notifications-versioning	✓	✓	✓	✓	✓		
RFC 8525 YANG Library	✓	✓	✓	✓	✓		✓
draft-ietf-netconf-yang-library-augmentation	✓	✓	✓	✓		✓	✓
RFC 9196 System and Notification Capabilities			✓	P	✓		✓
draft-netana-netconf-yp-transport-capabilities			✓	✓	✓		
RFC 9254 CBOR Named Identifiers	✓			✓	✓		

Green marked describes new capability at IETF 126. "P" to partially implemented.

Operational IETF/IEEE YANG

IETF 126 Module Implementation Status

	6WIND VSR	Huawei NE	Huawei MA	Cisco IOS XR	Arrcus Arcos
ietf-interfaces.yang (inventory, state and statistic)	✓	✓	✓		
ietf-hardware.yang (inventory, state and statistic)	✓	✓	✓		
ietf-alarms.yang (state)	✓	✓	✓		
ieee802-dot1ab-lldp.yang (inventory and state)	✓	✓			
ieee802-dot1ax.yang (inventory, state and statistic)					
ietf-bfd-ip-sh.yang (state)					
ietf-bfd-ip-mh.yang (state)					
ietf-bfd-lag.yang (state)					
ietf-isis.yang (state)					

Green marked describes new capability at IETF 126. "P" to partially implemented.

YANG Message Broker
IETF 126 Implementation Status

	NetGauze™	Pmacct	Apache Kafka	Ciena Blueplanet	Cisco Crosswork
draft-ietf-nmop-message-broker-telemetry-message	✓	✓			
YANG-Push Receiver	✓	✓			
YANG Schema Retrieval	✓				
YANG Message Broker Producer	✓				
YANG Schema Registry	na	na	✓	na	na
YANG Message Broker Consumer	✓			✓	✓
Validate Telemetry Message against YANG Schema	✓			P	P
draft-ietf-netmod-yang-anydata-validation	✓				
YANG Schema Comparison	na	na	✓	na	na

Green marked describes new capability at IETF 126. "P" to partially implemented.

YANG Library Feature Comparison

- An extensive YANG feature comparison between the different YANG library open-source implementations.
- Focuses on YANG features needed to validate YANG instance data against YANG schema and performing YANG to YANG transformations.

Table 15. Global synthesis of support levels across evaluated feature families

Feature family	libyang	Yangkit	ODL Yang Tools
Schema loading and validation	Supported	Supported	Partially supported
Module inspection	Partially supported	Supported	Supported
JSON data validation	Supported	Supported	Partially supported
CBOR data validation	Not supported	Supported	Not supported
Anydata validation	Supported	Partially supported	Partially supported
YANG structures	Supported	Supported	Not supported
Anydata inside structures	Supported	Partially supported	Not applicable
Structures inside anydata	Supported	Supported	Not applicable
XPath/data extraction	Supported	Partially supported	Partially supported
Subtree validation	Supported	Supported	Supported
YANG-to-YANG transformation	Supported	Supported for the described add/update cases	Supported for the described add/update cases
YANG Library loading	Supported	Supported	Supported
Yang Push validation	Supported	Supported	Not applicable
Telemetry validation	Supported	Supported	Not applicable
Notification envelope validation	Supported	Supported	Not applicable
Schema comparison	Supported through specific yanglint branch	Implementable	Implementable

https://github.com/network-analytics/yang_tests/blob/master/report_v2.pdf

Next Steps from IETF 125 Hackathon…

- Re-do YANG Message Broker Consumer tests on updated testbed with Ciena Blue Planet UAA Workflow Engine
- Complete IETF YANG module subscription tests with 6Wind VSR
- Compare YANG feature coverage between libyang, yangkit and yangtools
- Revert import only modules for augmented-by workaround
- Preserve cached YANG-Push subscription state across receiver reload
- Obtain YANG-Push subscription state from publisher if missing
- Support Netconf session queuing for schema retrieval
- Support libyang 5.4.9 in YANG-Push Receiver and Message Broker Consumer
- Message Key and Topic Name proof of concept

Next Steps for IETF 127 Hackathon…

- Support building RFC 8525 YANG Library in Java Message Broker Consumer
- Support draft-netana-nmop-yang-anydata-validation and RFC 8525 YANG library in yangkit
- Support YANG-Push Message Broker Topic Naming with dynamic Apache Kafka topic creation API

Thanks to…

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- Deepya Mandadi – Ciena
- Sivakumar Sundaravadivel – Ciena
- IJsbrand Wijnands – Arrcus
- Irfan Mohammad – Arrcus
- Benoit Claise - Everything OPS
- Paolo Lucente – Pmacct
- Vivekananda Boudia – INSA Lyon
- Alex Huang Feng – INSA Lyon/T-Systems
- Maxence Younsi – INSA Lyon/Swisscom
- Ashvanth Kumar Selvakumaran - Telus
- Leonardo Rodoni - Swisscom
- Ahmed Elhassany – Swisscom
- Thomas Graf – Swisscom

Want to know more then join us at…

- NMOP on **Monday 09:00 – 11:00** at the Grand Klimt Hall 2 for Message Broker, Message Key, BMP Telemetry Message document updates and hackathon related details.
- NETMOD on **Thursday 09:00 – 11:00** at the Grand Klimt Hall 1 for YANG Library feature comparison related details.